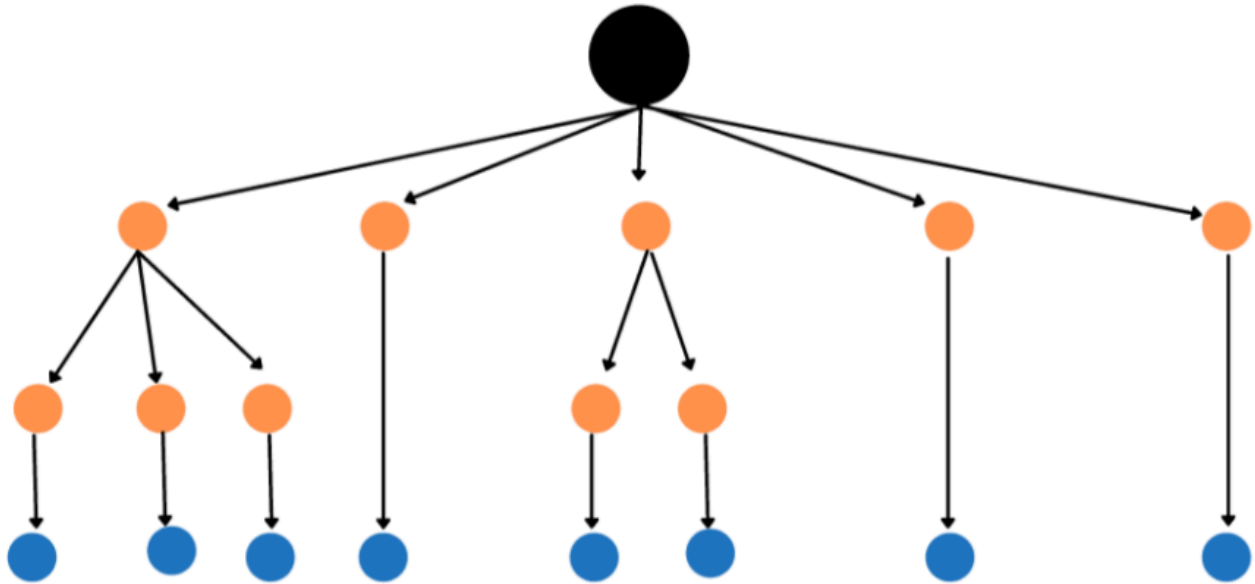


Decision Tree Model

Decision Tree



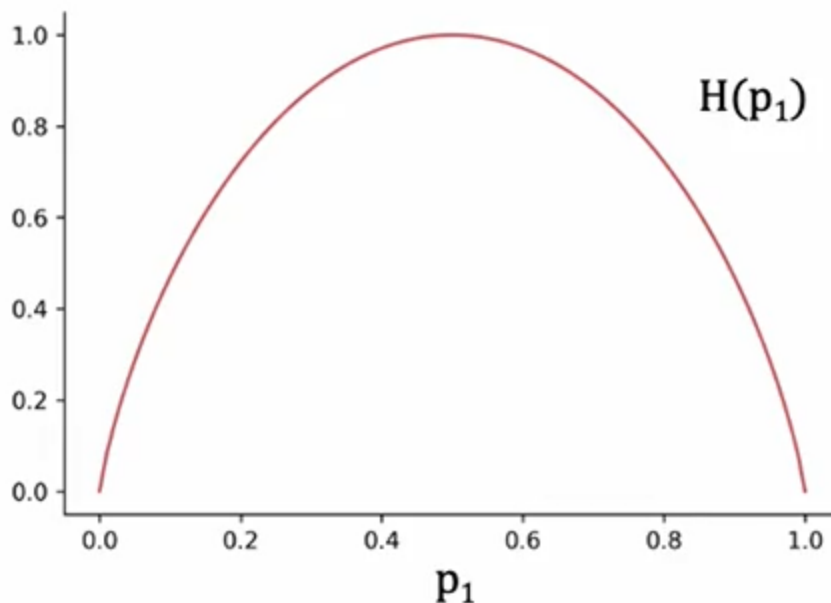
- Root Node, Decision Nodes, Leaf Nodes

Learning Process

- What Decisions to use?
 - Decisions which maximizes the purity
- When to stop splitting?
 - When the nodes reach a certain depth
 - When a node gets 100% result
 - When a node has less training example of desired result as compared to the other

Measuring Purity

- **Entropy** as a measure of Impurity



$$\begin{aligned}
 H(p_1) &= -p_1 \log_2(p_1) - p_0 \log_2(p_0) \\
 &= -p_1 \log_2(p_1) - (1 - p_1) \log_2(1 - p_1)
 \end{aligned}$$

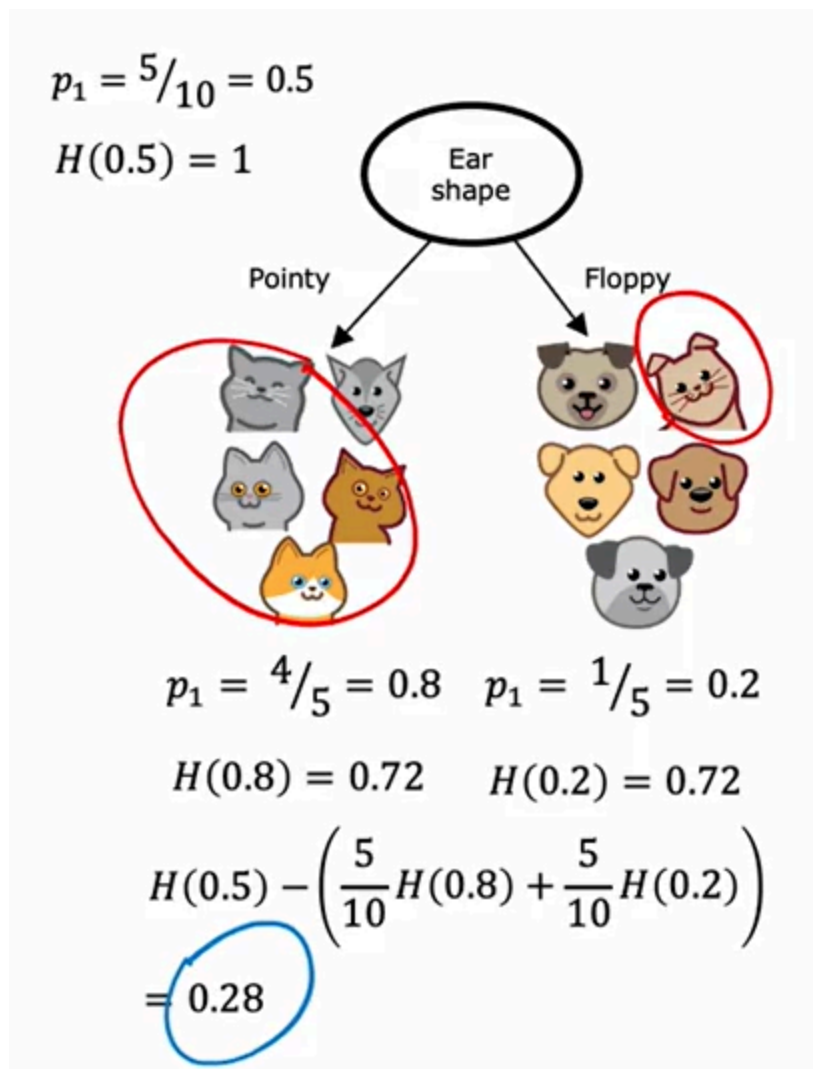
- p_x is the probability of positives in any node in a decision tree.

Choose a Split : Information Gain

- What set of features **reduces entropy** the most.
 - Reduction of entropy is called **Information Gain**.
- Information Gain
 - Taking a **Weighted Average** and subtracting it with the entropy of the previous node

Information gain

$$= H(p_1^{\text{root}}) - \left(w^{\text{left}} H(p_1^{\text{left}}) + w^{\text{right}} H(p_1^{\text{right}}) \right)$$



- The reduction in the entropy resulted in a split is called **information gain**
- The increase in information gain result in better model

Using one-hot encoding of Categorical Features

- Convert each category into new feature

Continuous Valued Features

- Change the threshold according to the value of information gain

Using Multiple Decision Trees

- Tree Ensemble
 - From the multiple decision tree we get multiple outputs. we consider the output which is repeated more times.

Sampling with Replacement

- Here we take a small sample not from the training example and create a new sample.
- Doesn't matter if the samples are repeated.
- With individual samples you can create a decision tree and get the output value from multiple of those and from that you can get the predicted value.

Random Forest Algorithm

Randomizing the feature choice

At each node, when choosing a feature to use to split, if n features are available, pick a random subset of $k < n$ features and allow the algorithm to only choose from that subset of features.

$$k = \sqrt{n}$$

Random forest algorithm

- Where does a Machine Learning Engineer go camping?
 - To a Random Forest

XGBoost

- Extreme Gradient Boosting
- Same like creating a Tree Ensemble but instead of picking from all examples with equal probability, make it more likely to pick misclassified examples from previously trained tree.

- Its one of the highly competitive algorithms in competitions.

Using XGBoost

Classification

```
from xgboost import XGBClassifier

model = XGBClassifier()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

When to use Decision Trees

- Works well in structured data and not preferred for unstructured data.
- Faster to train

When to use Neural Networks

- work on all types of data
- works with transfer learning
- slower to build